

CDS - 50.038

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INITIAL PROJECT PRESENTATION

PROJECT STATEMENT

Can audio features extracted from raw music files, combined with social engagement signals, predict a song's streaming success in a given region?

DATASET

Used Now:

- Kaggle Spotify Music Data: Spotify daily hit charts since beginning of 2017
- Manual Music Feature Extraction: `yt-dlp` to download mp3 samples of Spotify hit songs with `librosa` for feature extraction

Future Considerations:

- Google Trends: Search interest over time by region, available via `pytrends`
- Kaggle Billboard API / Data — Kaggle has scraped Billboard Hot 100 datasets going back decades.
- X API — Social media trends analysis through tweets

Visualizing Global Time-Series Data



Example Music Features Extracted

Loudness
(RMS Energy)



Tempo BPM



Brightness
(Spectral Centroid)

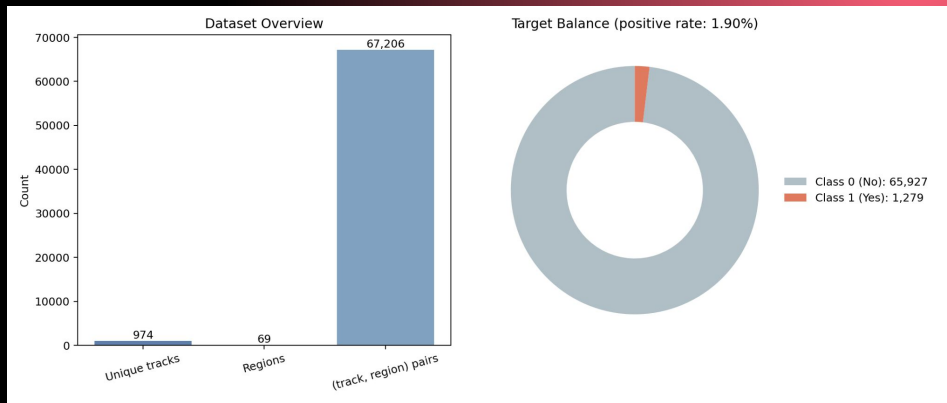


Timbre (MFCCs)



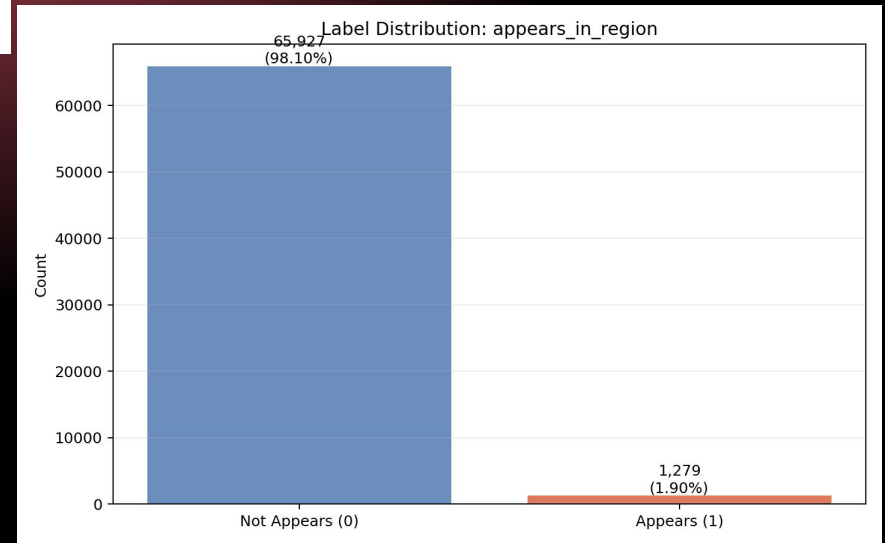
Percussiveness
(zero crossing rate)

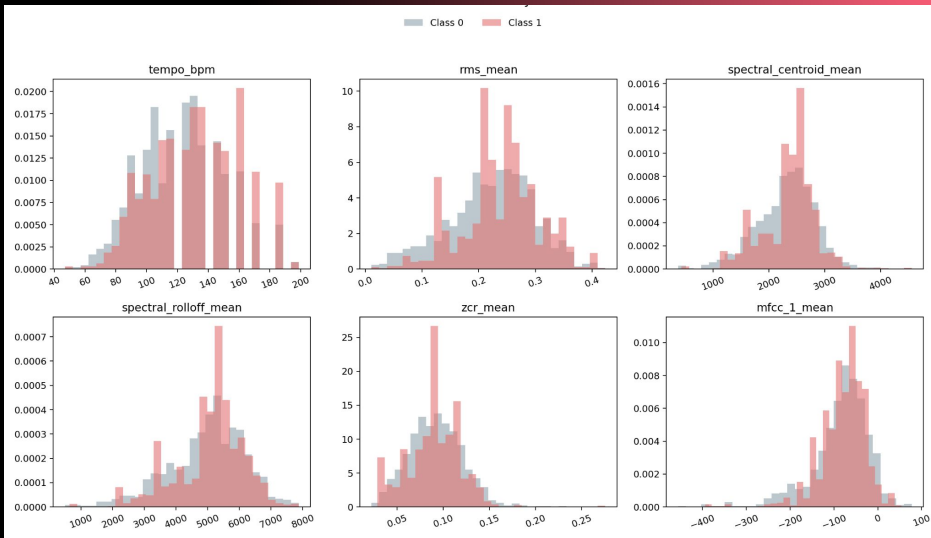




- Overall dataset of our model draws from is the third column— track-region pairs; it checks the virality of each song in a given region, allowing us to expand the dataset without having to download too many songs.
- Of the song-region pairings, 1.90% indicated virality, meaning that the song in that specific region will be viral, or will appear in the top 200 songs of the region.

- Since this percentage is so low, it indicates a heavy imbalance. This will require future fixing so the model will develop identification patterns on the viral songs rather than the majority non-viral songs.

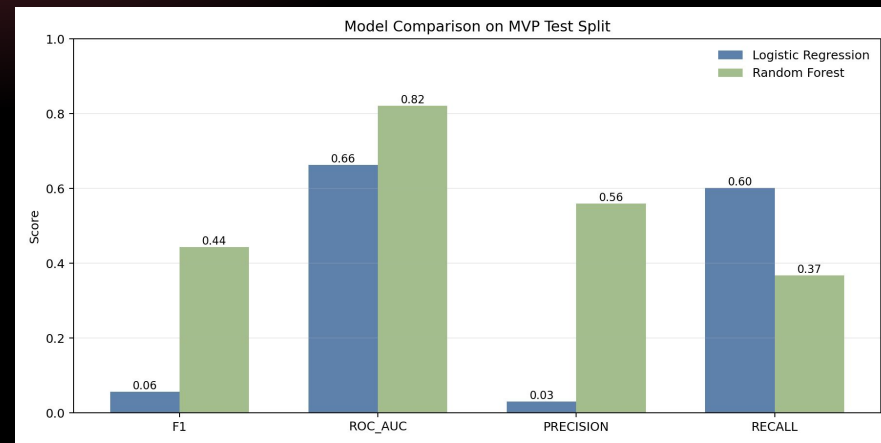




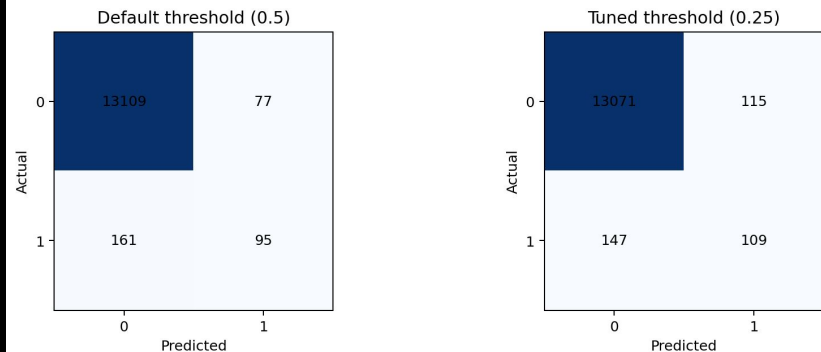
The graph on the right compares logistic regression model of the dataset versus a random forest model. Given the accuracy indicators along the x-axis, we can observe that random forest has a higher overall score for most metrics besides recall. This indicates that in general, random forest performs more accurately as a classification model. This can be attributed to the note that since our current model is binary, it cannot be properly regressed, as each song in a given region is either viral or not.

Features Extracted:

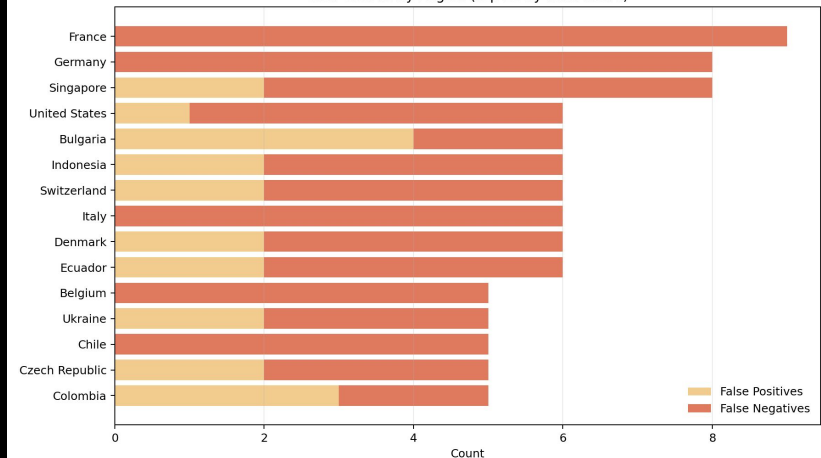
- **Tempo/BPM**: beats per minute of the song (speed of the song)
- **Modulation Spectrum (MS)**: measures change in amplitude and frequency. Used for genre detection and rhythm analysis.
- **MFCC**: Identifies timbre and musical “emotions.” Best for overall song feature analysis.
- Spectral Centroid
- Spectral Rolloff
- Zero-Cross Rate (ZCR)



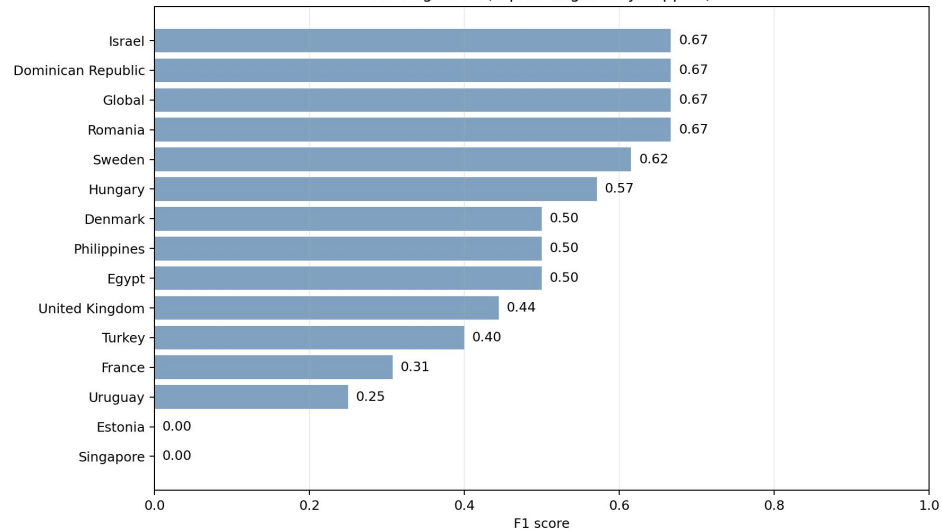
Confusion Matrices: Selected Model



Error Counts by Region (Top 15 by total errors)



Per-Region F1 (Top 15 Regions by Support)



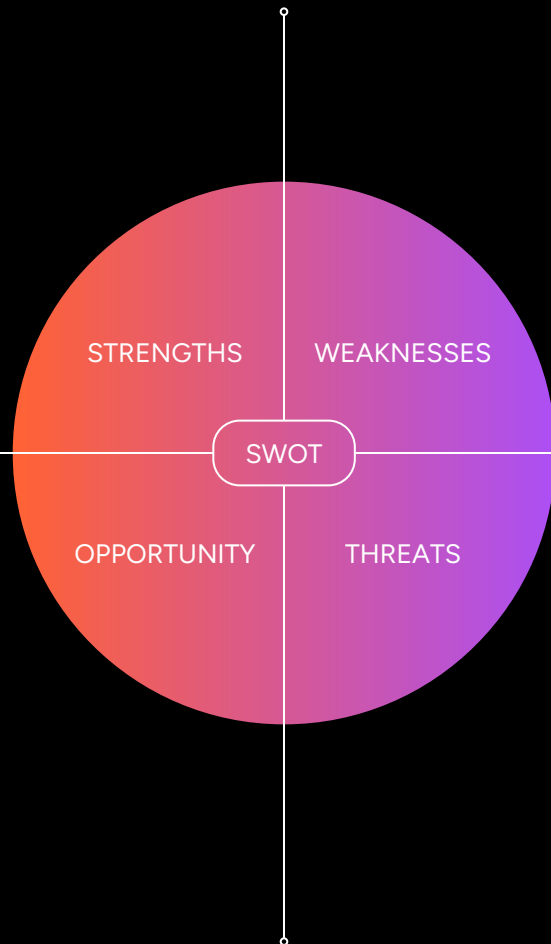
Confusion Matrices: Identifies the number of songs in the regions that were true positives or true negatives. The default graph has high precision but low recall. The tuned graph finds more true popular songs but makes more mistakes.

F1 per region: The mid to high range scores denote a well-performing model in those regions listed, while the low end scores shows a weaker model. This means the model does not perform equally everywhere.

MODEL COMPARISON

- Our model is **realistic**; music performs very different from region to region. A lower positive rate is to be expected.
- The dataset is **scalable**. The model will only perform better as more songs are added to the dataset.
- Derived from raw audio files. The model **avoids reliance on pre-existing streams**, and classifies the music based on the audio features themselves.

- Expand dataset and engineer more features to create a more developed and precise model.
- There is ability to expand beyond just regional music identification; we can also learn what makes a song universally viral and create broader generalizations.
- Factor in outside datasets such as Google trends or social media to increase virality identification factors.



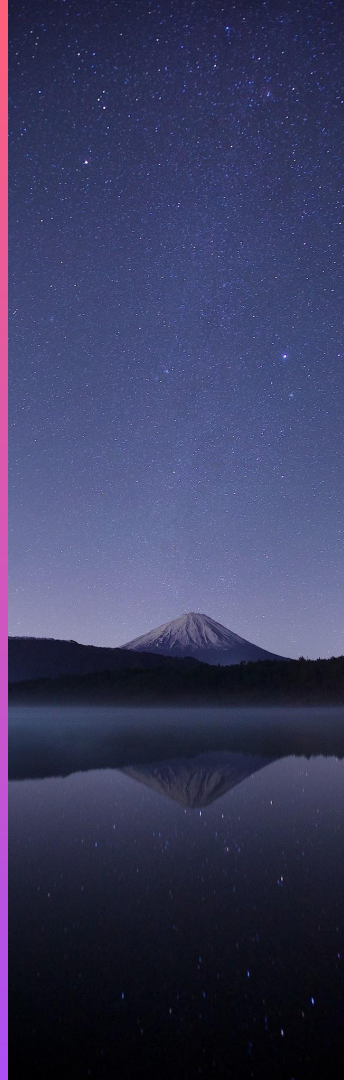
- Not full training set; We do not have the computational power to expand the set (i.e. manually extracting features from every song).
- Currently only doing **binary classification** (is viral or is not viral in a given region).
- Model mostly uses audio descriptors and regions to determine virality in an area, there are no outside virality factors (yet).
- Limitations on downloading music; it is preferable to use current music to develop the model, however there are limitations on downloading current music.
- Space and capacity; Downloading music takes a lot of storage space, and volume is bigger than what most average laptop spaces have.
- Add region features in order to balance the dataset more.

THANK YOU

Any questions?

IDENTIFY VALIDITY OF
RESULTS F1, ROC-AUC, ETC
AND EXPLAIN MEANING

Write a bold, compelling
statement about what
the company wants to
achieve



HOW WE'RE DOING

STRENGTHS

- Make a list of what your organization or team excels at
- This is what sets you apart from the competition
- Add as many items as you need

WEAKNESSES

- Identify areas for improvement within your organization or team
- Think about what you could be doing internally to achieve better results
- Add as many items as you need

OPPORTUNITIES

- Call out any opportunities for growth
- These are external factors, such as market trends or platforms, that your business can leverage
- Add as many items as you need

THREATS

- Mention any challenges, obstacles, or risks that your business is facing
- These can range from environmental factors to new market competitors
- Add as many items as you need

COMPETITOR ANALYSIS

	Competitor	Competitor	Competitor	Competitor	Competitor
Feature	✓		✓	✓	✓
Feature		✓			
Feature	✓	✓	✓		
Feature	✓	✓		✓	
Feature		✓	✓	✓	

OBJECTIVES

01

Objective title

Define your business, marketing, and communication objectives. For example, you might set goals around growing revenue, generating leads, or shifting perceptions around your brand.

02

Objective title

Each objective can be standalone or contribute to the next one in the list. This will depend on your strategic plan and internal company structure.

03

Objective title

Make sure your goals are specific, measurable, achievable, relevant, and time-bound (SMART). Explain what you want to achieve, for whom, during which time frame, and how you'll define success.

04

Objective title

For example, if you want to increase market share, aim for a specific percentage increase within the next year by launching a new product or capturing a new market segment.

Marketing objectives

Your marketing goals should be aimed at reaching, engaging, and capturing your target audience at every touchpoint, whether online or in-store.

- List 3 or 4 marketing objectives
- Make sure your goals are specific, measurable, achievable, relevant, and time-bound (SMART)
- For example, you might want to generate a specific number of leads next month through a targeted email campaign

Communication objectives

Your communication goals should be both internal and external. Think about brand awareness in the market, but also about employee engagement in the office.

- List 3 or 4 communication objectives
- Make sure your goals are specific, measurable, achievable, relevant, and time-bound (SMART)
- For example, you might want to increase web traffic a specific amount during the next quarter through SEO optimization

Business objectives

Your business goals are your overarching objectives. From annual revenue to market share and customer satisfaction, this is what everything contributes to.

- List 3 or 4 business objectives
- Make sure your goals are specific, measurable, achievable, relevant, and time-bound (SMART)
- For example, you might want to increase customer retention next year by automating and speeding up customer service



Persona name

Introduce your persona, who they are, and where they come from.

Your customer persona is a hypothetical person who represents an audience segment. Although the individual might be fictional, this information should be derived from your market research.

Needs and motivations

Describe what your persona wants to achieve. List their goals and outline their motivations to pursue those goals.

Pain points

Describe an obstacle your persona struggles with. Think about what interferes with their needs, goals, and motivations.

Opportunities

This is where your offering comes in. Think about how you can help this persona with their daily struggles and blockers.

- Call out specific opportunities as bullet points
- Each opportunity should reflect a feature, product, or service that you can provide
- Think about what you're not already offering: opportunities for growth, expansion, new launches, etc.



Persona name



What does your persona want, and what motivates them to pursue it?

- Age:** 00
- Occupation:** Job title or industry
- Status:** Marital status
- Education:** Degree
- Location:** Place name
- Audience:** Segment name

“Quote the persona on their needs and wants. This can be a fictional line or an excerpt from a real user interview.”

Pain points

- What prevents your persona from achieving their goals?
- Add 2 or 3 frustrations or blockers that are relevant to your products or services

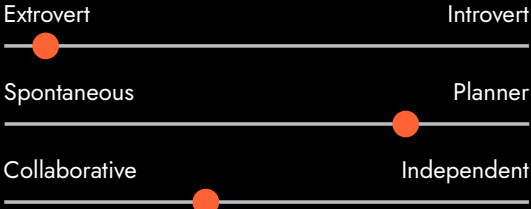
Opportunities

- What opportunities does this persona represent for your business?
- Add 2 or 3 opportunities that reflect a feature, product, or service that you can provide

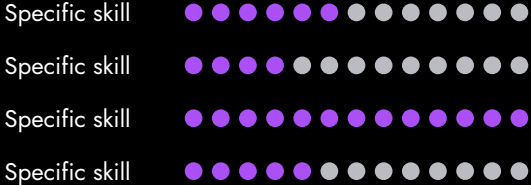
Favorite brands

LOGO	LOGO	LOGO
LOGO	LOGO	LOGO

Personality



Technical skills



Current feelings

Frustrated	Hopeful
Anxious	Enthusiastic

BUSINESS ROADMAP

What steps must we take to reach our strategic goals?

Step 01

This is a high-level summary. Focus on key milestones.

Step 02

For example, mention campaign or project kickoffs, product launches, etc.

Step 03

Add relevant dates, industry events, deliverables, etc.

Step 04

Add any goals or metrics you want to achieve within a certain time frame.

Step 05

Duplicate this slide if you want to include additional steps.

Step 06

BUSINESS ROADMAP

Step 01

What steps must we take to reach our strategic goals?

- Add more info about this step. For example, deliverables, activities, marketing assets
- You can also call out the teams or stakeholders involved
- Mention any performance targets that should be reached during this step

Step 02

What steps must we take to reach our strategic goals?

- Add more info about this step. For example, deliverables, activities, marketing assets
- You can also call out the teams or stakeholders involved
- Mention any performance targets that should be reached during this step

Step 03

What steps must we take to reach our strategic goals?

- Add more info about this step. For example, deliverables, activities, marketing assets
- You can also call out the teams or stakeholders involved
- Mention any performance targets that should be reached during this step

PROJECTED RESULTS

Make a claim about your growth outlook.

Business metrics

00%

Return on investment

\$00

Gross revenue

\$00

Net revenue

Customer metrics

00%

Customer retention

00%

Conversion rate

00

Site visits

PROJECTED RESULTS

Sales growth

A brief note on the results

00%

Return on
investment

A brief note on the results

\$00

Revenue

A brief note on the results

00K

TIMELINE

Show a timeline with the current status of the strategic plan and upcoming deadlines

We are here

Milestone 1

Quarter, Year

Outline how you'll achieve the strategic plan, step by step.

Milestone 2

Quarter, Year

Outline how you'll achieve the strategic plan, step by step.

Milestone 3

Quarter, Year

Outline how you'll achieve the strategic plan, step by step.

Milestone 4

Quarter, Year

Add as many milestones as you need. Duplicate this slide if necessary.

TIMELINE

Show a timeline with the current status of the strategic plan and upcoming deadlines



GANTT CHART

